Incident Report

Phishing-Based Ransomware Simulation in a Virtualized Environment

1. Introductory Summary

This incident report documents a simulated cybersecurity incident conducted as part of a Work Integrated Learning (WIL) project. The objective of the simulation was to demonstrate how a phishing attack can lead to ransomware-like behavior within a controlled laboratory environment. The incident involved the delivery of a phishing email to a test user, execution of a benign ransomware simulation script on a Windows virtual machine, and subsequent analysis of host and network artifacts. The simulation was performed ethically and safely, without causing permanent damage or data loss.

2. Incident Overview

This incident report documents a simulated cybersecurity incident conducted as part of a Work Integrated Learning (WIL) project. The objective of the simulation was to demonstrate how a phishing attack can lead to ransomware-like behavior within a controlled laboratory environment. The incident involved the delivery of a phishing email to a test user, execution of a benign ransomware simulation script on a Windows virtual machine, and subsequent analysis of host and network artifacts. The simulation was performed ethically and safely, without causing permanent damage or data loss.

3. Environment Description

The simulation was conducted within a VirtualBox-based virtual environment to isolate the project from the real world. The host machine had 8 GB RAM and limited storage, which influenced design decisions.

Virtual Machines Used:

- **Windows 10 Pro** – Victim Machine
- **Debian 12** – Utility Server
- **Kali Linux** – Attacker Machine

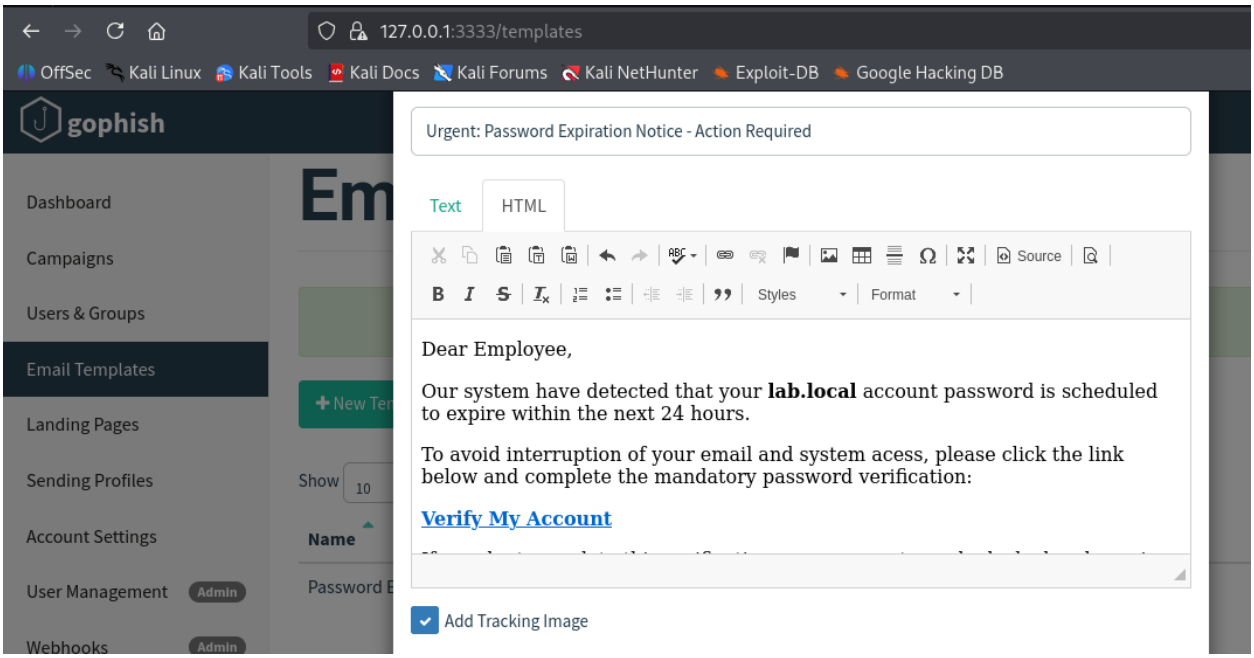
Notable Constraints:

- No Domain Controller or File Server was implemented
- Logical evidence acquisition was used instead of full disk imaging
- Some monitoring tools (Sysmon) provided limited output due to environment constraints.

4. Attack Vector and Execution

Phishing Attack

A phishing email was crafted and delivered using GoPhish. The email contained a deceptive message designed to entice the user to interact with a malicious link. Mail delivery was handled by Postfix, and mailbox access was managed via Dovecot on the Debian utility server.



Payload Execution

Instead of deploying real malware, a benign PowerShell script was used to simulate ransomware behavior. The script encrypted (locked) files within the user's Documents and Desktop directories and generated a ransom note. This approach ensured ethical compliance and prevented real harm.

```
PS C:\Users\testuser> powershell -ExecutionPolicy Bypass -File C:\Tools\simulate_ransom.ps1
Starting benign ransomware simulation...
[+] Locked: Doc1.rtf from C:\Users\testuser\Documents
[+] Locked: Doc2.odt from C:\Users\testuser\Documents
[+] Locked: Doc3.rtf from C:\Users\testuser\Documents
[+] Locked: Doc4.odt from C:\Users\testuser\Documents
[+] Locked: Doc5.rtf from C:\Users\testuser\Documents
[+] Locked: Image.bmp from C:\Users\testuser\Desktop
[+] Locked: Report.txt from C:\Users\testuser\Desktop
Simulation complete. Ransom notes created in Documents and Desktop.
PS C:\Users\testuser>
```

5. Evidence Collected

Windows (Victim)

- Locked files with modified extensions
- Ransom note created on the Desktop
- PowerShell execution output
- Windows Security and PowerShell event logs
- Sysmon logs (limited to Event ID 5)

Debian (Utility)

- Postfix logs confirming email delivery
- Dovecot logs confirming mailbox access
- Suricata logs (ICMP and baseline traffic)
- Packet capture (PCAP) files collected using tcpdump

6. Timeline Reconstruction

Time	Event	Source
14:35:12	Phishing email sent	Gophish
14:35:40	Email delivered	Postfix logs
19:27:03	Email accessed	Dovecot logs
23:55:52	Payload Executed	Powershell output
23:55:54	Files locked	Windows file system
23:55:05	Network traffic observed	PCAP

7. Detection and Monitoring Effectiveness

While Sysmon logging was partially functional, other logging mechanisms successfully captured critical stages of the incident. Windows Security logs, PowerShell output, and file system artifacts provided sufficient host-based evidence. Network monitoring through tcpdump and Suricata captured supporting traffic, although no direct ransomware signatures were triggered due to the benign nature of the payload.

```
Terminal
emily@Utility:~$ su
Password:
root@Utility:/home/emily# tcpdump -r week5_phishing_attack.pcap host 192.168.56.105
reading from file week5_phishing_attack.pcap, link-type EN10MB (Ethernet), snapshot length 262144
19:27:03.717287 ARP, Request who-has Utility tell 192.168.56.105, length 46
19:27:03.717713 ARP, Reply Utility is-at 08:00:27:c5:22:11 (oui Unknown), length 28
19:27:17.442188 ARP, Request who-has 192.168.56.110 tell 192.168.56.105, length 46
19:27:18.214218 ARP, Request who-has 192.168.56.110 tell 192.168.56.105, length 46
19:27:19.230889 ARP, Request who-has 192.168.56.110 tell 192.168.56.105, length 46
19:38:35.907990 IP 192.168.56.105 > igmp.mcast.net: igmp v3 report, 1 group record(s)
19:38:35.908010 IP 192.168.56.105 > igmp.mcast.net: igmp v3 report, 1 group record(s)
19:38:35.908011 IP 192.168.56.105.mdns > mdns.mcast.net.mdns: 0 ANY (QM)? DESKTOP-3RIPF7Q.local. (3
9)
19:38:35.908043 IP 192.168.56.105.mdns > mdns.mcast.net.mdns: 0 ANY (QM)? DESKTOP-3RIPF7Q.local. (3
9)
19:38:35.908068 IP 192.168.56.105.51756 > 224.0.0.252.5355: UDP, length 33
19:38:35.908071 IP 192.168.56.105.mdns > mdns.mcast.net.mdns: 0*- [0q] 2/0/0 AAAA fe80::9f8:2061:af
6f:1341, A 192.168.56.105 (77)
19:38:35.908077 IP 192.168.56.105.mdns > mdns.mcast.net.mdns: 0*- [0q] 2/0/0 AAAA fe80::9f8:2061:af
6f:1341, A 192.168.56.105 (77)
19:38:35.920085 IP 192.168.56.105 > igmp.mcast.net: igmp v3 report, 1 group record(s)
19:38:35.922847 IP 192.168.56.105 > igmp.mcast.net: igmp v3 report, 1 group record(s)
19:38:35.923865 IP 192.168.56.105.mdns > mdns.mcast.net.mdns: 0 ANY (QM)? DESKTOP-3RIPF7Q.local. (3
9)
```

```
Terminal
emily@Utility:~$ su
Password:
root@Utility:/home/emily# sudo tail -f /var/log/suricata/fast.log
12/24/2025-19:45:39.795632  [**] [1:9000001:1] ICMP Ping Detected [**] [Classifi
cation: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:d1c9:8a86:a5f8:a72
0:135 -> ff02:0000:0000:0000:0000:0001:ff9c:a246:0
```

8. Impact Assessments

The impact of the incident was limited to the test environment. Files within designated user directories were locked, simulating data unavailability typically associated with ransomware attacks. No permanent data loss occurred, and all changes were reversible through backups and snapshots.

9. Limitations

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10. Lessons Learned and Recommendations

This simulation demonstrated the effectiveness of phishing as an initial attack vector and highlighted the importance of user awareness, endpoint logging, and layered monitoring. Future improvements could include centralized log management, enhanced IDS tuning, and stronger endpoint detection controls. Utilizing a laptop with 16 GB RAM or greater would also improve the overall set up, execution and success of such an incident response simulation lab.

11. Conclusion

The simulated incident successfully demonstrated a phishing-initiated ransomware scenario within a controlled lab environment. Despite hardware and architectural limitations, sufficient evidence was collected to reconstruct the attack and analyze its impact. The project met its objectives and provided practical insight into incident response, detection, and forensic analysis.